NOCTURNAL AI · HIRING

AI Automation Engineer

Take-home assessment

Deadline: Sunday 11 October 2026, 23:59 PKT

Pick one task · about 6–8 focused hours

Welcome

Thank you for your interest in Nocturnal AI. We build AI systems inside real businesses (manufacturers, exporters, distributors) and keep running them after go-live.

This role sits next to senior executives. They describe a problem in a two-minute voice note, and you turn it into an automation they can rely on, quickly and safely.

This assessment is the job in miniature:

- pick **one** of four tasks
- build it
- send it to us

If it's strong, we'll invite you to present it in person.

How it works

- **Choose** one of the four tasks below.
- **Effort:** about 6–8 focused hours. Scope it down rather than overrun.
- **Deadline: Sunday 11 October 2026, 23:59 PKT.**
- **Questions:** by email until Friday 9 October. We reply within 24 hours.
- **Next:** we review every submission by Tuesday 13 October. Shortlisted candidates present in person later that week.

The company (fictional)

Kestrel Industries is a Karachi manufacturer and exporter: about 900 staff, two factories, and buyers in Europe, the UK and the US. Its directors run the business on WhatsApp and email. They're short on time, they don't log in to systems, and they've been burned before by automations that broke quietly.

Everything about Kestrel is invented: the people, phone numbers, emails, documents and figures. The phone numbers are deliberately invalid, so send every test message to yourself.

Scenario clock: treat "now" as **Thursday 15 October 2026, 08:30 PKT**. Make it a setting, so we can replay your automation at that moment.

Choose one task

Each task starts with the request as the director sent it: a voice note, transcribed. The data is in the attached pack (`Nocturnal-Assessment-Data.zip`).

Like real company data, it is messy on purpose, and we won't tell you where. Finding the problems and handling them sensibly is part of the task.

Task 1 · Approvals chaser

> **Director Finance:** "Purchase and payment approvals get stuck with people for days, and nobody knows who's sitting on what. Chase them for me. Remind the approver, and if they still don't act, go to their boss. Anything five lakh and above needs my approval too. Every evening, send me one message: what got approved, what's stuck, and with whom. And I should be able to approve from my phone without logging in to anything."

Data:

- `t1_requests.csv`
- `t1_people.csv`
- `t1_policy.md`: the rules (working hours, reminder and escalation timings, the director threshold).

Must have:

- Reminders and escalations that follow the policy, with no duplicate nagging.
- Approve or reject from the phone (one tap or one reply), with the request updated.
- An audit log of every reminder, escalation and decision: who, what, when, and through which channel.
- The 18:00 summary to the Director Finance.

Nice to have:

- Delegation when an approver is on leave.
- The director can ask "what's stuck with Hina?" and get an answer.

Task 2 · Director's morning brief

> **Managing Director:** "Every morning before nine I want one WhatsApp: yesterday's dispatches and the pending payments. Short. Tell me what's wrong first. And don't make up numbers. If the system and the gate don't agree, tell me."

Data:

- `t2_dispatches_erp.csv`: the ERP's dispatch register.
- `t2_gate_log.csv`: security gate exits.
- `t2_receivables.csv`: open invoices.
- `t2_fx_rates.csv`.
- `t2_md_calendar.ics`: the MD's day.

Must have:

- One message at a fixed time, readable in 30 seconds on a phone.
- Exceptions first: dispatches where the ERP and the gate disagree, and payments that need the MD's attention.
- Every figure traceable to a source row. If an LLM writes the text, code checks the numbers.
- If a source is missing or late, the brief still goes out and says what's missing.

Nice to have: the MD replies "3" and gets the detail behind item 3.

Task 3 · Document desk

> **CEO's office:** "Every few months someone's passport or visa expires right before a trip, and we're scrambling. Track all of them, warn people early, and warn me if anyone is about to travel on a passport that won't be valid. These are passports, so be careful with them."

Data:

- `t3_documents.csv`: staff travel documents and company certificates, all fake.
- `t3_trips.csv`: upcoming trips.
- `t3_rules.md`: alert timings, travel rules and privacy rules.

Must have:

- Alerts at 90, 30 and 7 days to the holder and their PA, each sent once.
- A travel check that flags problems before the trip.
- The weekly digest for the admin office.
- Document numbers masked in every message and log, and a data store that isn't open to everyone.

Nice to have:

- A "renewal in progress" status that pauses reminders.
- A photo of a new passport, emailed in, with the expiry read automatically and confirmed by a person before it's saved.

Task 4 · Register reconciliation

> **Finance Controller:** "Our store keeps a handwritten receiving register, the ERP has its own entries, and the storekeeper keeps their own Excel. They never match, and month-end takes three days. I want to photograph the register pages and have the system tell me exactly where the three disagree, and prove nothing has been missed."

Data:

- `t4_register_page1.jpg` and `t4_register_page2.jpg`: photos of the September register.
- `t4_erp_grn.csv`: the ERP's goods-received notes.
- `t4_store_sheet.xlsx`: the storekeeper's sheet.

Must have:

- The register photos turned into structured rows. Anything unclear is flagged for a person, not guessed.
- A three-way reconciliation:
  - matched
  - mismatched (which field, which values)
  - missing from each source
- Proof of completeness: every row of every source lands in exactly one outcome, and the counts and totals tie back to the inputs.
- A review step where a person resolves only the doubtful items, not everything.

Nice to have:

- A monthly run that only processes new pages.
- A short summary message for the controller.

Task 4 is the most data-heavy. It suits you if you lean towards AI engineering; Tasks 1–3 suit you if you lean towards automation. All four are equally welcome.

Ground rules

- **Use only our data.** Never use real personal data, including your own documents.
- **Any stack.** Use whatever you'd really use at work: n8n, Make, Zapier, Power Automate, Python, TypeScript, Apps Script. Free tiers are enough; don't pay for anything.
- **Messaging.** WhatsApp is the directors' channel. Use one of:
  - the official WhatsApp Cloud API test number
  - the Twilio WhatsApp sandbox
  - Telegram, Slack or email, with a note on how you'd move it to WhatsApp

  Don't use unofficial WhatsApp Web libraries.
- **No secrets in the submission.** Keep API keys, tokens and passwords out of the repository, screenshots and video.
- **AI tools are welcome** (Claude Code, Codex, Cursor, ChatGPT and so on). Tell us which ones you used, and for what. In the in-person session you'll change your own build live, so make sure you understand all of it.
- **Small and working beats big and half-working.** Write down what you'd build next instead of building it.

What to send

Reply to the email this came with, by **Sunday 11 October 2026, 23:59 PKT**:

1. **The build.**
   - A GitHub repo link (public, or private and shared with `sarmad-nocturnal`), or a zip.
   - For n8n, Make or Zapier: the exported workflow file, plus screenshots of the setup.
2. **A demo video, 3–5 minutes** (Loom, unlisted YouTube or Google Drive). Run it end to end on our data, and show one failure and how your automation handles it.
3. **A README, two pages at most:**
   - what it does, and how to run it

    - the assumptions you made
    - how it fails, and how someone would know
    - how you handled the data
    - what you'd do with another week
    - which AI tools you used
4. **A rough time log:** hours spent, by part.

How we'll evaluate

Roughly in order of weight:

1. **Would a busy director use it?** Short, exceptions first, the right channel at the right time.
2. **Is it right?** Figures trace back to the source; the data's problems are found and handled, not hidden.
3. **Does it work end to end** from your README, on our data?
4. **Does it fail safely?** Retries, no duplicate messages, and someone hears about it when it breaks.
5. **Data handling:** least access, masked identifiers, no secrets in code.
6. **How clearly you explain it:** the README, the video, and the questions you ask us.

We don't score which tool you chose, visual polish, or the number of features.

The in-person session (shortlisted candidates)

About 45 minutes, in Karachi, during the week of 12 October. We'll confirm a slot with you.

- **10 min:** you demo your automation.
- **15 min:** one of us plays the director. They use it, then change a requirement, and you make a small change live.
- **10 min:** what happens when it breaks, and how the data is handled.
- **10 min:** your questions for us.

Questions

Reply to the same email. We answer within 24 hours until Friday 9 October. We share every answer with all candidates, without names.

Good questions count in your favour: asking the right ones is most of this job.